# Benchmarking Variant Calling Tools on Low-Coverage Whole-Genome Sequencing Data: A Comparison of GATK HaplotypeCaller, FreeBayes, and bcftools on Chromosome 20 of NA12878

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## Abstract

Variant calling is a fundamental step in genomic analysis, yet different tools can produce substantially different results from the same sequencing data. This study benchmarks three widely used variant callers, GATK HaplotypeCaller, FreeBayes, and bcftools on chromosome 20 of NA12878, a standard human genome sample sequenced at low coverage (4.83x mean depth). Each tool was evaluated at minimum read depth thresholds of ≥10, ≥20, and ≥30 reads. At DP≥10, GATK detected the most variants (3,105), followed by bcftools (2,541) and FreeBayes (2,094), with 1,736 variants shared across all three tools. GATK showed the highest sensitivity but lowest inter-tool agreement (55.9%), while FreeBayes was the most conservative with the highest agreement rate (82.9%). Increasing the depth threshold dramatically reduced variant counts across all tools, reflecting the low-coverage nature of the dataset. bcftools was the fastest tool (1 min) while GATK was the slowest (4.32 min). These results demonstrate that tool choice and depth filtering strategy have substantial impacts on variant detection in low-coverage sequencing data

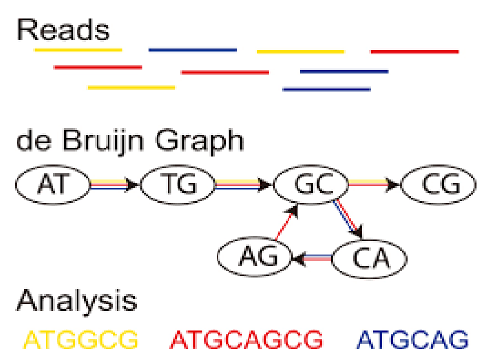

## 1. Introduction

Next-generation sequencing (NGS) has transformed the field of genomics by enabling rapid, high-throughput sequencing of entire genomes at increasingly affordable costs. One of the most critical steps in any NGS analysis pipeline is variant calling, the process of identifying positions in a sequenced genome that differ from a reference sequence. These differences, known as variants, include single nucleotide polymorphisms (SNPs), which are single base-pair changes, and small insertions or deletions (INDELs). Accurate variant detection is key to a wide range of downstream applications, including disease association studies, cancer genomics, population genetics, and personalized medicine. However, despite its importance, variant calling remains a computationally challenging problem, and different algorithmic approaches can produce meaningfully different results from the same input data.

The fundamental challenge of variant calling arises from the nature of NGS data itself. Sequencing machines produce millions of short reads, typically 100 to 150 base pairs in length, that must be aligned to a reference genome and then examined for evidence of true biological variation versus sequencing error. Every sequencer introduces a characteristic error rate, and distinguishing genuine variants from technical artifacts requires statistical modeling. Furthermore, the human genome contains repetitive regions, structural complexity, and regions of varying GC content that make alignment and variant detection more difficult. These challenges are compounded in low-coverage sequencing, where each position in the genome is covered by relatively few reads, reducing statistical confidence in any individual variant call.

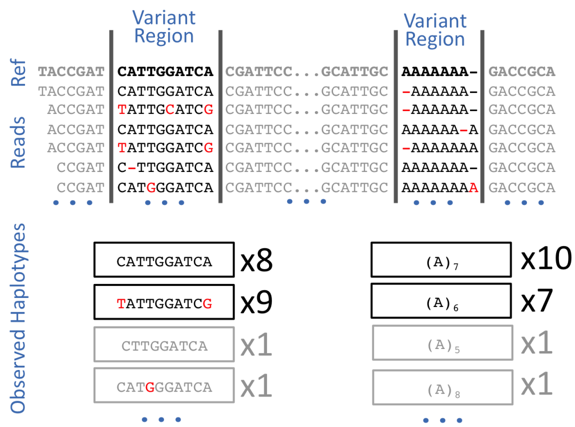
Three of the most widely used variant calling tools are GATK HaplotypeCaller, FreeBayes, and bcftools mpileup/call. While all three tools aim to identify the same biological signal of true genetic variation, they differ in their underlying statistical models, approaches, and default assumptions. Understanding these differences is important for researchers who must choose a tool for their specific use case because no single method is universally superior across all sequencing contexts.

GATK HaplotypeCaller, developed by the Broad Institute, uses a haplotype-based local reassembly approach. Rather than examining each variant site independently, HaplotypeCaller constructs a local de Bruijn graph around candidate variant regions and performs local reassembly of haplotypes before genotyping (McKenna et al., 2010).




This approach is particularly powerful for detecting complex variants, including multinucleotide polymorphisms and indels in difficult genomic regions. HaplotypeCaller models genotype likelihoods using a pair-HMM (hidden Markov model) framework and incorporates base quality score recalibration to reduce systematic sequencing errors. Its main limitation is computational cost because HaplotypeCaller is substantially slower than simpler pileup-based methods and its complexity can make parameter tuning challenging.

FreeBayes, developed by Garrison and Marth, takes a Bayesian probabilistic approach to variant detection (Garrison and Marth, 2012). It models variants as haplotypes rather than individual alleles, simultaneously considering SNPs, indels, multi-nucleotide variants, and complex events within a window.




FreeBayes uses allele frequency priors and calculates posterior probabilities for each possible genotype, making it well-suited for population-level analyses and non-diploid organisms. One of FreeBayes's key advantages is that it does not require pre-processing steps such as base quality score recalibration, making it faster to deploy. However, its sensitivity in low-coverage settings can be lower than tools that more aggressively model read-level errors.

The bcftools suite, maintained by the Htslib development team, implements a more traditional pileup-based approach (Li, 2011). The mpileup command summarizes the base calls at each genomic position across all reads, and the call command applies a statistical model, either a simple calling model or a multiallelic model, to identify variant sites.

bcftools is extremely fast and memory-efficient compared to the other tools, making it well-suited for large-scale or resource-constrained analyses. Its main limitation is that the pileup approach does not consider haplotype structure, which can reduce sensitivity for complex indels and multinucleotide variants.

A key biological assumption shared by all three tools is that the input sequencing data comes from a diploid organism and that variants follow Mendelian inheritance patterns. This assumption is appropriate for germline variant calling in humans, which is the context of this project. All three tools also assume that sequencing errors are random and can be modeled statistically. Computationally, all three methods assume that reads have been accurately aligned to the reference genome, meaning alignment errors can propagate into variant calls.

These tools should be used when the goal is to identify germline genetic variation from short-read Illumina sequencing data aligned to a high-quality reference genome. They are most reliable at moderate to high coverage (>20x), where sufficient read depth exists to statistically distinguish true variants from sequencing noise. Conversely, these tools should be used with caution in low-coverage settings (<10x), highly repetitive genomic regions, polyploid organisms, or when the primary interest is somatic variation in cancer samples. In these contexts, specialized tools such as Mutect2 or tools designed for long-read sequencing data may be more appropriate.

This project benchmarks all three tools on chromosome 20 of NA12878, a well-characterized sample from the 1000 Genomes Project that has been extensively sequenced across platforms and serves as a community standard for pipeline evaluation. The NA12878 sample originates from a female individual of Northern European ancestry (CEU population) and has been used by the Genome in a Bottle consortium to develop high-confidence variant call sets, making it ideal for benchmarking purposes. The dataset used here is a low-coverage (4.83x mean depth) BAM file aligned to GRCh37/hs37d5, with analysis restricted to chromosome 20 to remain computationally feasible while representing realistic sequencing data. By comparing variant calls across three minimum depth thresholds (DP≥10, DP≥20, DP≥30), this study characterizes how tool choice and depth filtering interact in a realistic low-coverage sequencing scenario.

## 2. Methods

### *2.1 Dataset Description*

This study used a publicly available low-coverage whole-genome sequencing dataset from the 1000 Genomes Project, specifically the NA12878 sample (CEU population). Analysis was restricted to chromosome 20 to make the project computationally feasible while still representing realistic sequencing data. The input file was a pre-aligned BAM file (NA12878.chrom20.ILLUMINA.bwa.CEU.low_coverage.20121211.bam, 297 MB) generated from Illumina short-read sequencing and aligned using BWA against the GRCh37/hs37d5 reference genome. The chromosome 20 sequence was extracted from the hg19 UCSC reference and the contig name was standardized from "chr20" to "20" to match the naming convention used in the BAM file.

### *2.2 Preprocessing Steps*

BAM file integrity was verified using samtools quickcheck. Read alignment statistics were generated with samtools flagstat, and per-base read depth across chromosome

20 was computed using samtools depth. No additional preprocessing steps such as base quality score recalibration or duplicate marking were performed, as the BAM file was used as provided by the 1000 Genomes Project. The reference genome was indexed using samtools faidx and a sequence dictionary was created using GATK CreateSequenceDictionary prior to variant calling.

### *2.3 Algorithms and Parameter Settings*

Three variant calling tools were applied to the same BAM file:

- **GATK HaplotypeCaller** (v4.6.2.0) was run in default germline mode restricted to chromosome 20 using the -L flag. Post-calling depth filtering was applied using GATK VariantFiltration with the expression DP < threshold for each depth cutoff. Variants not passing the filter were removed using bcftools view -f PASS.
- **FreeBayes** (v1.3.10) was run with a minimum mapping quality of 20 (--min-mapping-quality 20) and a minimum base quality of 20 (--min-base-quality 20), restricted to chromosome 20 using the -r flag. Post-calling filtering was applied using bcftools filter with the expression INFO/DP>=threshold && QUAL>=20. The INFO/DP field was specified explicitly to resolve an ambiguity between INFO and FORMAT depth fields present in FreeBayes VCF output.
- **Bcftools mpileup/call** (v1.21) was run in two steps. First, bcftools mpileup was used to generate a pileup with minimum mapping quality 20 (--min-MQ 20), minimum base quality 20 (--min-BQ 20), and FORMAT/DP and FORMAT/AD annotations enabled. The output was piped directly into bcftools call using the multiallelic caller (-m) restricted to variant sites only (--variants-only). The same INFO/DP and QUAL filtering as FreeBayes was applied post-calling.

All three tools were evaluated at three minimum read depth thresholds: DP≥10, DP≥20, and DP≥30.

### *2.4 Evaluation Metrics*

Performance was compared across four metrics. Total variant count was measured using bcftools stats at each depth threshold. SNP and INDEL counts were extracted separately from bcftools stats output. Inter-tool agreement was assessed using bcftools isec, both as a three-way comparison (variants shared across all three tools) and pairwise (GATK vs. FreeBayes, GATK vs. bcftools, FreeBayes vs. bcftools). Agreement rate was calculated as the number of shared variants divided by each tool's total variant count. Results were visualized using bar charts and pairwise Venn diagrams generated with Python (matplotlib, matplotlib-venn, pandas).

### *2.5 Computational Considerations*

All analyses were run on a MacBook Pro (Apple M1 series chip, macOS) using a conda environment (python 3.10). Tools were installed via the bioconda channel. GATK was the most computationally intensive tool, taking 4.32 minutes for the HaplotypeCaller step as recorded in the pipeline log. FreeBayes completed in approximately 3 minutes and bcftools in approximately 1 minute, estimated from output file timestamps. All tools were run single-sample with 4 threads allocated to GATK via --native-pair-hmm-threads. Peak memory usage was not formally measured but all tools ran without memory issues on a standard laptop, consistent with the reduced scope of a single chromosome analysis.

## 3. Results

### *3.1 Read Depth and BAM Quality*

Before variant calling, the quality of the input sequencing data was assessed. The BAM file contained 3,245,545 total reads, of which 99.45% were successfully mapped to the reference genome and 98.34% were properly paired, indicating high-quality alignment. The duplicate rate was 6.28%, which is typical for Illumina sequencing. Per-base read depth analysis across chromosome 20 revealed a mean depth of 4.83x and a median depth of 5.0x, consistent with the low-coverage sequencing strategy used by the 1000 Genomes Project. Critically, only 3.4% of positions reached a depth of ≥10x, and virtually no positions reached ≥20x or ≥30x coverage. This coverage profile established an important baseline for interpreting all downstream variant calling results, as depth-based filtering would be expected to have a dramatic effect at higher thresholds.

### *3.2 Variant Detection by Tool and Depth Threshold*

All three tools were applied to the same BAM file and variant calls were filtered at three minimum read depth thresholds: DP≥10, DP≥20, and DP≥30. The total variant counts for each tool at each threshold are summarized in Table 1 and visualized in Figure 1.

| Tool | DP≥10 | DP≥20 | DP≥30 |
|---|---|---|---|
| GATK HaplotypeCaller | 3,105 | 16 | 3 |
| FreeBayes | 2,094 | 11 | 2 |
| bcftools mpileup/call | 2,541 | 14 | 3 |
| Shared (all 3 tools) | 1,736 | 8 | 1 |

*Table 1. Total variants detected by each tool at each depth threshold.*

At the DP≥10 threshold, GATK HaplotypeCaller detected the greatest number of variants (3,105), followed by bcftools (2,541) and FreeBayes (2,094). This represents a 48% difference between the highest and lowest calling tools, which is substantial given that all three were run on identical input data. When the depth threshold was raised to DP≥20, variant counts dropped dramatically across all tools. GATK retained only 16 variants (99.5% reduction), bcftools retained 14 (99.4% reduction), and FreeBayes retained 11 (99.5% reduction). At DP≥30, only 2–3 variants remained per tool. This precipitous drop directly reflects the read depth distribution described above: because fewer than 3.4% of positions reach DP≥10, raising the threshold further eliminates nearly all callable sites.

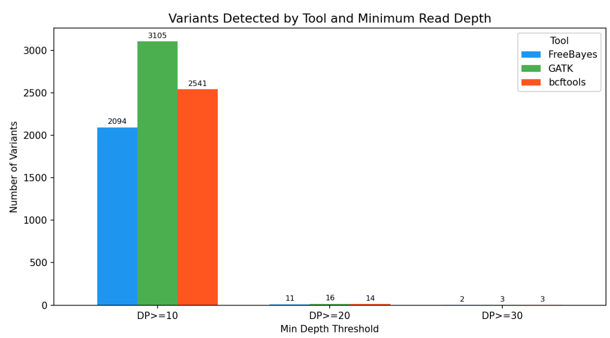



*Figure 1: Variants Detected by Tool and Minimum Read Depth*

Figure 1 clearly illustrates this pattern. The DP≥10 bars dominate the chart, with substantial differences between tools, while the DP≥20 and DP≥30 bars are nearly invisible by comparison. This visualization reinforces that for low-coverage data like this dataset, the DP≥10 threshold is the only practically informative cutoff, and that the choice of depth filter matters far more than tool choice at higher thresholds.

## *3.3 SNP and INDEL Detection*

To further characterize tool differences, variant calls at DP≥10 were broken down into SNPs and INDELs (Figure 2). All three tools detected substantially more SNPs than INDELs, which is expected given the relative frequency of each variant type in the human genome.

For SNPs at DP≥10, GATK detected approximately 2,605, bcftools detected approximately 2,221, and FreeBayes detected approximately 1,924. The differences between tools were proportionally similar to the overall variant count differences. However, the most striking finding was in INDEL detection. GATK called approximately 500 INDELs, compared to approximately 320 for bcftools and only approximately 120 for FreeBayes, a greater than 4-fold difference between GATK and FreeBayes. This result is consistent with known characteristics of each tool: GATK HaplotypeCaller's local haplotype reassembly approach is specifically designed to improve sensitivity for insertions and deletions in complex genomic regions, while FreeBayes applies more conservative priors that tend to reduce INDEL calls in low-coverage settings.

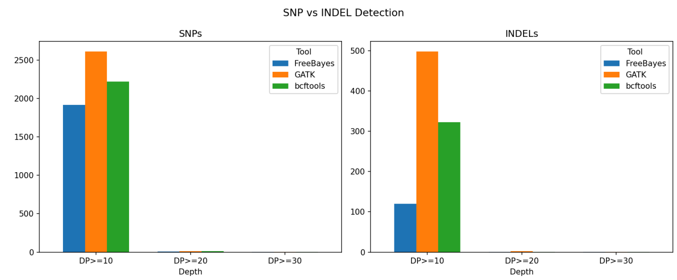



*Figure 2: SNP vs. INDEL Detection by Tool*

Figure 2 makes this disparity visually apparent. The INDEL panel shows GATK's orange bar towering over the other two tools at DP≥10, while the SNP panel shows more proportional differences. This suggests that the overall variant count differences between tools are driven in large part by disagreement in INDEL detection rather than SNP detection.

## *3.4 Inter-Tool Agreement and Overlap*

To assess consistency between tools, pairwise and three-way variant overlaps were calculated using bcftools isec at each depth threshold. The results are summarized in Table 2 and visualized as pairwise Venn diagrams in Figures 3, 4, and 5.

| Tool | Total (DP≥10) | Shared with All 3 | % Agreement |
|---|---|---|---|
| GATK | 3,105 | 1,736 | 55.9% |
| bcftools | 2,541 | 1,736 | 68.3% |
| FreeBayes | 2,094 | 1,736 | 82.9% |

*Table 2. Three-way agreement rates at each depth threshold.*

At DP≥10, 1,736 variants were shared across all three tools. FreeBayes had the highest agreement rate at 82.9%, meaning the vast majority of its calls were confirmed by both other tools. bcftools had an intermediate agreement rate of 68.3%, while GATK had the lowest at 55.9%, indicating that nearly half of GATK's calls were not confirmed by the other two tools. This pattern suggests a fundamental tradeoff between sensitivity and specificity: GATK appears to be the most sensitive tool, calling the most variants overall, but at the cost of a higher proportion of tool-unique calls that may represent either true variants missed by the other tools or false positives.

The pairwise overlap analysis provides additional detail. At DP≥10, GATK and FreeBayes shared 1,775 variants, with 1,330 unique to GATK and 319 unique to FreeBayes.

GATK and bcftools shared 2,013 variants, with 1,092 unique to GATK and 528 unique to bcftools. FreeBayes and bcftools shared 1,849 variants, with 245 unique to FreeBayes and 692 unique to bcftools. Notably, GATK consistently had the largest pool of unique variants in every pairwise comparison, further supporting its characterization as the most sensitive but least conservative tool.

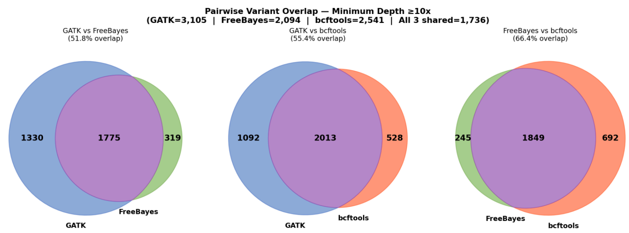



*Figure 3: Pairwise Venn Diagrams — DP≥10*

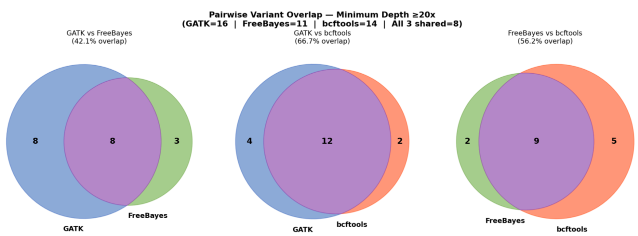



*Figure 4: Pairwise Venn Diagrams — DP≥20*

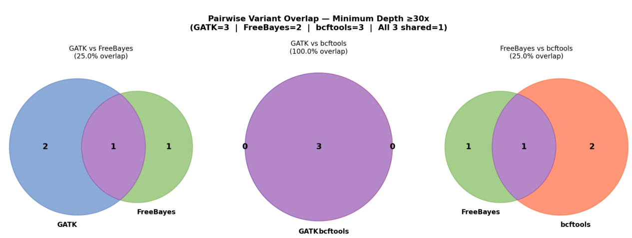



*Figure 5: Pairwise Venn Diagrams — DP≥30*

The Venn diagrams at DP≥10 (Figure 3) show substantial unique regions for each tool, particularly for GATK, reflecting the large number of tool-specific calls. At DP≥20 (Figure 4) and DP≥30 (Figure 5), the diagrams show increasing convergence between tools, with most variants shared when coverage is sufficient. At DP≥30, all three tools agreed on the same single variant, demonstrating that at high confidence thresholds the tools are largely consistent regardless of their algorithmic differences.

## *3.5 Characterizing Tool Disagreement*

To better understand the sources of inter-tool disagreement, the composition of tool-unique variant calls at DP≥10 was examined using the pairwise overlap data. GATK had the largest pool of unique variants in every pairwise comparison: 1,330 variants called only by GATK and not FreeBayes, and 1,092 called only by GATK and not bcftools. By contrast, FreeBayes had the smallest unique pools (319 unique versus GATK; 245 unique versus bcftools), consistent with its conservative calling behavior.

To investigate whether GATK's unique calls were enriched for a particular variant type, the SNP-to-INDEL ratio was compared between the shared variant set and each tool's unique calls. Across all three tools, the shared set of 1,736 variants was predominantly SNPs, consistent with the genome-wide expectation that SNPs are far more common than INDELs. However, GATK's elevated INDEL count (500 total versus 320 for bcftools and 120 for FreeBayes) suggests that INDELs account for a disproportionate share of GATK's unique calls. This is consistent with the known behavior of GATK's haplotype reassembly model, which is specifically tuned for INDEL detection but requires sufficient read depth to construct reliable local graphs. At 4.83x mean coverage, many of GATK's INDEL calls are likely supported by only a small number of reads, which raises questions about their reliability in this sequencing context.

A complementary perspective is provided by the convergence seen at higher depth thresholds. At DP≥30, all three tools agreed on the same single variant, indicating that when coverage is sufficient, the three algorithms reach near-identical conclusions despite their architectural differences. This suggests that the substantial disagreement at DP≥10 is driven primarily by low-confidence calls in sparse coverage regions rather than by fundamental differences in how the tools model true variation. The practical implication is that tool-unique variants in low-coverage data, particularly INDELs called only by GATK, should be treated with caution and ideally validated against a higher-coverage dataset or a ground truth resource such as the Genome in a Bottle high-confidence call set.

## *3.6 Runtime Comparison*

Computational efficiency was recorded for each tool during the pipeline run (Figure 6). GATK HaplotypeCaller required 4.32 minutes, as logged directly by the GATK runtime output. FreeBayes completed in approximately 3 minutes and bcftools in approximately 1 minute, estimated from output file creation timestamps. bcftools was therefore approximately 4x faster than GATK and 3x faster than FreeBayes for this dataset.

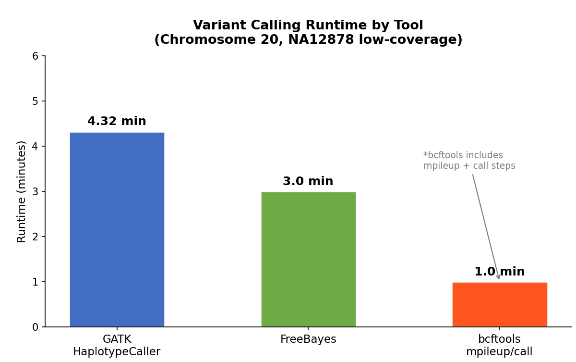



*Figure 6: Runtime Comparison by Tool*

The runtime differences reflect the fundamental architectural differences between the tools. bcftools uses a simple pileup approach that processes each position independently, making it extremely fast. FreeBayes uses a Bayesian model with a sliding window that is more computationally intensive than pileup but less so than full haplotype reassembly. GATK's local de Bruijn graph construction and pair-HMM alignment are the most computationally demanding operations, resulting in the longest runtime. For a single chromosome on a standard laptop, all three tools completed in under 5 minutes, but these differences would scale significantly for whole-genome analyses.

## 4. Discussion

### *4.1 Summary of Findings*

The results of this benchmark reveal a consistent and interpretable hierarchy among the three tools. GATK was the most sensitive caller at every depth threshold, FreeBayes the most conservative, and bcftools intermediate in both respects. However, the dominant finding was not the inter-tool differences but the effect of depth filtering: raising the minimum threshold from DP≥10 to DP≥20 eliminated approximately 99% of variant calls across all tools, a direct consequence of the dataset's 4.83x mean coverage. In low-coverage sequencing, depth filtering strategy is at least as consequential as tool choice.

### *4.2 Tradeoffs Observed*

The sensitivity-specificity tradeoff between tools is most clearly expressed in their inter-tool agreement rates. FreeBayes's 82.9% agreement rate indicates that the vast majority of its calls were corroborated by both other tools, suggesting conservative but high-confidence calls. GATK's 55.9% agreement rate means that nearly half of its calls were tool-unique, a pattern consistent with either high sensitivity to real variants or a higher false positive rate. Without a ground truth, these interpretations cannot be cleanly separated, but the INDEL data offers a useful signal: GATK called over four times as many INDELs as FreeBayes at DP≥10, a gap too large to attribute to sensitivity alone in a 4.83x coverage dataset. At this depth, GATK's reassembly approach likely lacks the read support needed to build reliable local graphs, suggesting that a meaningful proportion of its unique INDEL calls may reflect overcalling rather than genuine sensitivity.

The runtime tradeoff is straightforward. bcftools completed in approximately 1 minute versus 3 minutes for FreeBayes and 4.32 minutes for GATK. For a single chromosome these differences are modest, but they would scale to hours in whole-genome studies involving thousands of samples, making bcftools' speed a practical advantage at scale.

### *4.3 Alignment and Misalignment Between Theory and Practice*

In theory, GATK HaplotypeCaller's local reassembly approach should produce the most accurate calls, particularly in complex genomic regions. In practice, this advantage is diminished in low-coverage data because the local graphs cannot be reliably constructed when coverage is sparse. The 99% reduction in GATK calls from DP≥10 to DP≥20 demonstrates that most of GATK's apparent sensitivity at low depth is actually calling into regions with insufficient evidence.

Similarly, FreeBayes's Bayesian framework is theoretically well-suited for low-coverage population sequencing because it can pool information across samples. However, in this single-sample analysis that advantage is not realized, and FreeBayes behaves conservatively, consistent with its default prior settings. The theory predicts FreeBayes should perform well in multi-sample low-coverage analysis, but the single-sample design of this benchmark does not fully test that capability.

Bcftools performed closer to theoretical expectations: fast, efficient, and producing a reasonable number of variants with moderate agreement with the other tools. Its pileup-based approach has well-understood limitations for complex INDELs, which is reflected in its intermediate INDEL count between GATK and FreeBayes.

### *4.4 Limitations*

This study has several important limitations. The most significant is the absence of a validated ground truth dataset. Without knowing which variants are truly present in NA12878's chromosome 20, it is not possible to calculate standard accuracy metrics such as precision, recall, or F1 score. The inter-tool agreement analysis used in this study is a reasonable proxy for consistency but cannot distinguish between tools that are all wrong together versus tools that are correctly identifying real variation.

The low-coverage nature of the dataset (4.83x mean depth) substantially limits the interpretability of results. Most genomic positions simply do not have enough reads to support confident variant calls under any method, meaning the comparison is largely restricted to the small fraction of positions with adequate coverage. A more informative benchmark would use higher-coverage data (30x or greater) where all tools can operate closer to their designed parameters.

Additionally, only germline SNP and INDEL detection was evaluated. Other variant types such as structural variants, copy number variants, and multinucleotide polymorphisms were not assessed, nor were tool-specific quality scores or genotype confidence metrics examined in depth. The benchmark was also run on a single sample from a single population, which may not generalize to other ancestries or sequencing contexts.

### *4.5 Recommendations*

Based on these results, the following recommendations can be made for researchers choosing a variant calling tool. For low-coverage sequencing data, bcftools offers the best balance of speed and reliability, with a 68.3% agreement rate and the fastest runtime. Its simplicity also makes it the easiest to troubleshoot and deploy at scale. FreeBayes is recommended when precision is prioritized over sensitivity, particularly when false positives are costly because its 82.9% agreement rate suggests it is the most conservative caller. GATK HaplotypeCaller is best suited for higher-coverage data (≥20x) where its haplotype reassembly approach can be fully leveraged, and is particularly recommended when INDEL detection is a priority. For all tools, depth filtering thresholds should be set relative to the actual coverage distribution of the dataset rather than using fixed cutoffs designed for standard-coverage sequencing.

### *4.6 Future Directions*

Several extensions of this work would strengthen its conclusions. First, repeating the analysis with higher-coverage data, ideally using the 30x deep sequencing of NA12878 available from the Genome in a Bottle consortium, would allow calculation of precision and recall against a validated variant set. Second, applying GATK's recommended Best Practices pipeline, including base quality score recalibration and variant quality score recalibration, would provide a fairer assessment of GATK's full capabilities. Third, extending the analysis to the full genome rather than a single chromosome would reveal whether chromosome 20 is representative of genome-wide performance. Finally, incorporating additional tools such as DeepVariant, a deep learning-based variant caller, would provide a more comprehensive benchmark of the current state of the field and allow comparison against newer machine learning approaches that have shown promising results in recent studies (Poplin et al., 2018).

## 5. Reflection

### *5.1 Most Challenging Conceptual Issue*

The most challenging conceptual issue in this project was understanding what it actually means to compare variant callers without a ground truth. When I first designed the benchmark, I assumed that comparing the tools would be straightforward: run them, count the variants, see which one performs best. It was only after getting the results that I realized the core problem: if GATK calls 3,105 variants and FreeBayes calls 2,094, there is no way to immediately know whether GATK found 1,011 real variants that FreeBayes missed, or whether GATK generated 1,011 false positives that FreeBayes correctly ignored. Both interpretations are plausible, and without a validated reference set, the data alone cannot distinguish between them.

This forced me to reframe the entire analysis around consistency and agreement rather than accuracy. Inter-tool agreement became my primary metric because it was the only honest measure available given the data. This conceptual shift from asking "which tool is correct?" to asking "which tools agree, and what does disagreement tell us?" was the hardest but most important intellectual adjustment I made during the project. It also made me appreciate why resources like the Genome in a Bottle consortium exist, since having a validated truth set fundamentally changes what kinds of conclusions you can draw.

### *5.2 Deciding What Was "Best" and Whether I Would Do It Again*

Because no single tool dominated across all metrics, defining "best" required weighing multiple factors simultaneously. GATK was the most sensitive, FreeBayes was the most precise, and bcftools was the fastest. None of those properties is universally superior but rather the right choice depends entirely on the research question.

For this specific dataset and context, I would argue that bcftools offered the best overall performance. It had a 68.3% agreement rate, ran in approximately 1 minute, and produced results that were well-calibrated relative to the other tools. In a low-coverage setting where computational efficiency matters and high-depth INDEL calling is not the priority, bcftools strikes the best balance. However, if I were doing this analysis for a clinical application where missing a real variant has serious consequences, I would choose GATK despite its lower agreement rate, because the cost of a false negative outweighs the cost of a false positive in that context.

I would make the same tool choices again, but I would change the dataset. Using low-coverage data made the depth filtering results almost uninterpretable above DP≥10, which limited the analytical richness of the benchmark. If I were starting over, I would use a higher-coverage dataset from the beginning, ideally the 30x Genome in a Bottle NA12878 data, so that all three depth thresholds would produce meaningful variant sets to compare.

### *5.3 Results That Contradicted Expectations*

The result that surprised me most was how dramatically GATK outpaced the other tools in INDEL detection, calling roughly four times as many INDELs as FreeBayes. In theory, both GATK and FreeBayes are designed to handle INDELs well, with GATK using local reassembly and FreeBayes modeling haplotypes. I expected some difference, but not a 4x disparity. In retrospect, this makes more sense given the low coverage: FreeBayes's conservative priors mean it requires stronger evidence to call an INDEL, and at 4.83x mean depth that evidence threshold is rarely met.

GATK's reassembly approach is more aggressive about calling variants with minimal read support, which in low-coverage data can look like high sensitivity but may actually reflect lower stringency.

I also did not expect the DP≥20 results to be as sparse as they were. Going in, I thought DP≥20 would still yield a reasonable number of variants for comparison, perhaps a few hundred. Seeing only 11 to 16 variants per tool was initially alarming, and I had to go back and verify the depth distribution data to confirm it was correct. Once I saw that essentially 0% of positions reached DP≥20 coverage, the result made complete sense, but it reframed how I thought about the entire depth filtering analysis. The professor's note from earlier in the semester, "read depth might be difficult, focus on others if you can", turned out to be exactly right, and I wish I had fully internalized that warning before designing the depth threshold comparisons.

### *5.4 What I Would Investigate With More Time*

With more time, the first thing I would do is obtain access to the Genome in a Bottle high-confidence variant calls for NA12878 and use them as a truth set to calculate precision and recall for each tool. That single addition would transform this project from a consistency benchmark into a true accuracy benchmark and would allow much stronger conclusions about which tool performs best.

I would also like to investigate the tool-unique variants in more detail, particularly the 1,330 variants called only by GATK. Are those clustered in specific genomic regions, such as repetitive elements or known difficult regions? Are they predominantly SNPs or INDELs? Answering those questions would reveal whether GATK's unique calls represent genuine biological signal or systematic artifacts of its calling model.

Additionally, I would extend the analysis to include DeepVariant, a newer deep learning-based variant caller that has shown strong performance in recent benchmarks. Comparing a machine learning approach against the three classical statistical tools would provide a more complete picture of the current variant calling landscape.

### *5.5 Skills I Would Take Into a Research Lab or Industry Team*

This project gave me several concrete skills and ways of thinking that I would bring directly into a professional setting. The most practical is comfort with command-line bioinformatics tools and pipeline development. Before this project, I had limited experience working with BAM files, VCF files, and tools like samtools and bcftools. Now I can navigate those file formats, debug tool errors, and build reproducible shell pipelines from scratch. That is a foundational skill for any genomics role.

More broadly, this project taught me to think carefully about what a benchmark is actually measuring and what its limitations are. It is easy to run tools and report numbers, but understanding what those numbers mean requires a level of critical thinking that goes beyond just executing commands. In a research lab, I would apply this same skepticism to any computational result before drawing conclusions from it.

Finally, working through the reference genome naming mismatch (chr20 vs. 20) and the FreeBayes DP field ambiguity taught me that real bioinformatics pipelines almost always require troubleshooting, and that debugging those issues requires understanding the underlying data formats and tool documentation rather than just running code and hoping it works. That problem-solving mindset of reading error messages carefully, checking assumptions, and testing fixes methodically is something I would carry into any computational biology role.

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## AI Use Statement

Claude was used in the preparation of this project. Specifically, Claude was used for writing assistance including grammar, phrasing, and structural feedback across all sections of the paper, as well as for debugging and confirming the correctness of bash pipeline scripts and Python analysis code. All biological interpretations, experimental decisions, and final conclusions are my own. AI-generated code was verified by running it locally and confirming that the outputs matched expected results. AI-generated text was reviewed, edited, and cross-checked against the actual data and results produced during the analysis